

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1717

Portage Pass Tap 138 kV
May 18, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1717. The Interconnection Customer proposes a 199.6 MW battery energy storage facility interconnecting at the Portage Pass Tap 138 kV substation in the Pennsylvania area. The request was received on April 11, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 380 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$67,683,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Flannigan-W E Swoope 138 KV, the Dinuba Energy-Menton 138 KV Line loads to 171.4% of its 156 MVA rating. Project DFAX: 100.0%.

For the loss of W E Swoope-Flannigan 138 KV, the Tuller Hill-Dinuba Energy 138 KV loads to 142.4% of its 195 MVA rating. Project DFAX: 100.0%.

Beebee Island Hydro Plant-Cliff Drive 138 KV: 135.5% loading (234 MVA) following the outage of the Flannigan-W E Swoope 138 KV; DFAX = 100.0%.

Tuller Hill-Edenvale Tap 138 KV: 135.5% loading (218 MVA) following the outage of the Flannigan-W E Swoope 138 KV Line; DFAX = 100.0%.

For the loss of W E Swoope-Flannigan 138 KV line, the Indian Harbour-Cliff Drive Substation 138 KV Line loads to 134.7% of its 246 MVA rating. Project DFAX: 100.0%.

The Beebee Island Hydro Plant Substation-Edenvale Tap 138 KV is overloaded to 134.1% of its 231 MVA emergency rating for the loss of the Flannigan Substation-W E Swoope Substation 138 KV Line. The project contributes 100.0% DFAX (199.6 MW) to this overload.

The W E Swoope-Portage Pass Tap 138 KV is overloaded to 132.5% of its 230 MVA emergency rating for the loss of the Cliff Drive-Indian Harbour 138 KV. The project contributes 100.0% DFAX (199.6 MW) to this overload.

The Flannigan-W E Swoope 138 KV Line is overloaded to 131.5% of its 252 MVA emergency rating for the loss of the Indian Harbour-Cliff Drive 138 KV L. The project contributes 100.0% DFAX (199.6 MW) to this overload.

Portage Pass Tap-Menton 138 KV: 125.6% loading (201 MVA) following the outage of the Flannigan-W E Swoope Substation 138 KV; DFAX = 100.0%.

Lost Hillschevon-Gray 138 KV: 113.7% loading (177 MVA) following the outage of the Flannigan 345/138kV Auto; DFAX = 75.6%.

The Panther Creek Energy Facility-Indian Harbour 138 KV is overloaded to 112.8% of its 326 MVA emergency rating for the loss of the Kline Township Cogen Facility 345-138 kV XFMR. The project contributes 12.0% DFAX (23.9 MW) to this overload.

The Redondo-Leake 138 KV is overloaded to 106.5% of its 225 MVA emergency rating for the loss of the Ontelaunee Energy Center-Eliza Creek 345 KV Line. The project contributes 7.7% DFAX (15.5 MW) to this overload.

Plaquemine-Lost Hillschevron 138 KV: 105.3% loading (182 MVA) following the outage of the Flannigan 345 kV/138 KV XFMR; DFAX = 75.6%.

The Canyon-Gonzales 138 KV is overloaded to 100.2% of its 260 MVA emergency rating for the loss of the Wigwam-Upsco 345 KV L. The project contributes 5.6% DFAX (11.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dinuba Energy-Menton 138 KV Line	Rebuild line with higher-capacity conductor	\$3,873,000	100% (\$3,873,000)
Tuller Hill-Dinuba Energy 138 KV	Rebuild line with higher-capacity conductor	\$4,933,000	100% (\$4,933,000)
Cliff Drive-Beebe Island Hydro Plant 138 KV	Rebuild line with higher-capacity conductor	\$3,684,000	100% (\$3,684,000)
Tuller Hill-Edenvale Tap 138 KV	Rebuild line with higher-capacity conductor	\$21,410,000	100% (\$21,410,000)
Indian Harbour-Cliff Drive Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$2,996,000	100% (\$2,996,000)
Beebe Island Hydro Plant-Edenvale Tap 138 KV Line	Rebuild line with higher-capacity conductor	\$4,074,000	100% (\$4,074,000)
W E Swoope-Portage Pass Tap 138 KV	Rebuild line with higher-capacity conductor	\$2,809,000	100% (\$2,809,000)
Flannigan-W E Swoope 138 KV Line	Rebuild line with higher-capacity conductor	\$1,231,000	100% (\$1,231,000)
Portage Pass Tap-Menton 138 KV	Rebuild line with higher-capacity conductor	\$11,432,000	100% (\$11,432,000)
Gray-Lost Hillschevron Substation 138 KV L	Rebuild line with higher-capacity conductor	\$4,555,000	100% (\$4,555,000)
Panther Creek Energy Facility-Indian Harbour 138 KV L	Rebuild line with higher-capacity conductor	\$4,666,000	44% (\$2,077,000)
Rdondo-Leake 138 KV	Reconductor line	\$655,000	100% (\$655,000)
Plaquemine-Lost Hillschevron 138 KV	Reconductor line	\$728,000	100% (\$728,000)
Canyon-Gonzales 138 KV	Reconductor line	\$3,226,000	100% (\$3,226,000)
	Total allocated to project		\$67,683,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.